

Session 3 · Deck 3.B5

Hands-On: From a Closed Form to a Calibrated Economy

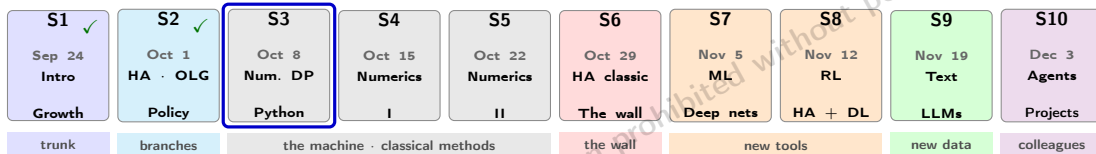
Zhigang Feng

Thursday, October 8, 2026

Build it on paper → solve it on the machine → push it to the frontier with AI

Where We Are

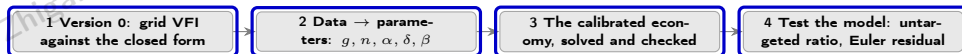
you are here: 3.B5 ▼



Build it on paper → solve it on the machine → push it to the frontier with AI

Block B: 3.B1 your toolkit · 3.B2 from loops to arrays · 3.B3 PyTorch, and coding with AI · 3.B4 the data workflow · 3.B5 hands-on: build it yourself.

Hands-on, 3.B5:



The Job: Four Steps

By the end of class you should have **your own** solver for a growth model calibrated to the U.S., and a reason to believe it.

1. **Version 0.** Grid VFI on the one model with a closed form. Grade it against the formula.
2. **Data $\rightarrow$ parameters.** Download seven series; turn them into $g, n, \alpha, \delta, \beta$.
3. **The calibrated economy.** Add growth to Step 1's code and raise γ from 1 to 2; solve at Step 2's numbers; check the steady state.
4. **Test the model.** A ratio you did not target, and the Euler residual.

The skill is the **sequence**: a case you can check, then one change at a time, each with a check.

	Today: class economy	HW1
frequency	annual	quarterly
utility	CRRA, $\gamma = 2$	CRRA, $\gamma = 1$ (log)
growth	g, n from data	none
unit test	$\alpha = 0.33, \beta = 0.96$	$\alpha = 0.36, \beta = 0.95$
data	PWT ratios, 1960–2019	yours (A3)

How to Work

- **Work in pairs.** One types, one checks; swap at each step.
- **Use an AI assistant if you like.** Give it the step, not the whole job — and remember 3.B3: validation is the part you cannot delegate.
- **A step is done when its checkpoint passes,** not when the code runs. Do not start the next step before.
- **Checkpoints are numbers with a tolerance.** Your data numbers may differ in the third digit if FRED has revised a series since the vintage on the slide.

What you need open

Python with `numpy` and `pandas` (L00; Colab works too) · a browser · the decks 3.A2–3.A4, 3.B2, 3.B4.

Exit ticket

At the end of class, hand in your pair's checkpoint numbers — the last step you reached, and the numbers you got. Unfinished steps are fine; wrong numbers you did not check are not.

Step 1: Version 0, Where You Know the Answer

Utility is CRRA throughout: $u(c) = c^{1-\gamma}/(1-\gamma)$, and $u(c) = \ln c$ at $\gamma = 1$ (its limit, up to a constant). Step 1 is $\gamma = 1$: $\max \sum_t \beta^t \ln c_t$ s.t. $c_t + k_{t+1} = k_t^\alpha$ (full depreciation), with $\alpha = 0.33$, $\beta = 0.96$. The answer is known:

$$g(k) = \alpha\beta k^\alpha, \quad k^* = (\alpha\beta)^{1/(1-\alpha)} = 0.1798.$$

1. A grid of $N = 500$ points on $[0.5k^*, 1.5k^*]$.
2. The return matrix $u(c_{ij})$, $c_{ij} = k_i^\alpha - k_j$, set to -10^{10} where $c_{ij} \leq 0$ (a number, not $-\infty$: 3.A4) — built **once**, by broadcasting.
3. From $V = 0$, iterate $V_i \leftarrow \max_j \{u_{ij} + \beta V_j\}$ until the largest change is below 10^{-6} ; count the sweeps.
4. Read the policy off the argmax . Report $\max_i |g(k_i) - \alpha\beta k_i^\alpha|/h$: the error in grid steps h .

Checkpoint 1

339 sweeps, counting the one that meets the tolerance; the largest policy error is **less than one grid step** — here 0.61 of a step.

Usual pitfalls

$\ln$ of a negative number gives nan, and nan poisons $\max$: mask infeasible choices *before* taking logs. A solver whose choices live on the grid cannot beat $h/2$.

Technique: 3.A2 (the algorithm), 3.A4 Steps 2–5, 3.B2 (broadcasting).

Step 2: Data → Parameters

Annual data from FRED, sample 1960–2019:

GDPCA, B23ORCOA052NBEA	real GDP; population (BEA)
LABSHPUSA156NRUG	labour share (PWT)
RKNANUSA666NRUG	capital, real (PWT)
RGDPNAUSA666NRUG	GDP, real (PWT)
CSHICPUSA156NRUG	investment share I/Y (PWT)
CSHCCPUSA156NRUG	consumption share C/Y (PWT)

1. $g = e^b - 1$, with b the slope of a least-squares line through $\ln(Y/N)$ against the year; n likewise from $\ln N$.
2. $\alpha = 1 -$ mean labour share. K/Y and $I/K = (I/Y)/(K/Y)$: compute each year, then average.
3. δ from capital accumulation, β from the Euler equation, with $\gamma = 2$ (3.B4 derives both):

$$\delta = \frac{I}{K} - [(1+g)(1+n) - 1], \quad \beta = \frac{(1+g)^\gamma}{1 + \alpha/(K/Y) - \delta}.$$

Checkpoint 2 (data of 2026-10-04)

g	1.95%	K/Y	3.54
n	1.04%	δ	0.038
α	0.384	β	0.971

Usual pitfalls

Index by **year** and slice with `.loc[1960:2019]` (inclusive); plain `[...]` on integers can slice by position. Growth rates are decimals: 0.0195, not 1.95.

Technique: 3.B4 (Acquire; the *Analyze* frames; Calibration: The Recipe).

How We Take Care of Growth

K , Y and C grow, so they have no steady state for a grid to hold. In the growing economy, A grows at rate g and N at rate n ,

$$Y_t = K_t^\alpha (A_t N_t)^{1-\alpha}, \quad C_t + K_{t+1} = Y_t + (1 - \delta)K_t,$$

and the household maximizes $\sum_t \beta^t N_t u(C_t/N_t)$. Divide by $A_t N_t$ — a hat means *per effective worker*.

$$\hat{c}_t + (1 + g)(1 + n) \hat{k}_{t+1} = \hat{k}_t^\alpha + (1 - \delta) \hat{k}_t,$$

$$N_t u(C_t/N_t) \propto [(1 + n)(1 + g)^{1-\gamma}]^t u(\hat{c}_t).$$

The bracket joins β^t : $\hat{\beta} = \beta(1 + n)(1 + g)^{1-\gamma}$ (next frame). Nothing else depends on t .

Population-weighted utility as in Cooley & Prescott (1995), "Economic Growth and Business Cycles".

The two new numbers

$G = (1 + g)(1 + n)$: a unit of $\hat{k}'$ costs G today — tomorrow there are more workers, each more productive, to equip.

$\hat{\beta}$: more people in the future, $(1 + n)$, each richer, $(1 + g)^{1-\gamma}$ — which counts for less when $\gamma > 1$. Need $\hat{\beta} < 1$.

Steady state = balanced growth path

$\hat{k}$ constant: K , Y , C grow by the factor G , output per person at rate g , every ratio is constant — so Step 2 targets **ratios**, and g is the trend of Y/N . The steady-state Euler equation, $(1 + g)^\gamma = \beta(\alpha \hat{k}^{\alpha-1} + 1 - \delta)$, is 3.B4's second equation.

Where $\hat{\beta}$ Comes From, and Why the Utility Is CRRA

Three facts:

- consumption per person is $c_t = C_t/N_t = A_t \hat{c}_t$;
- $N_t = N_0(1+n)^t$ and $A_t = A_0(1+g)^t$;
- CRRA is homogeneous: $u(A\hat{c}) = \frac{(A\hat{c})^{1-\gamma}}{1-\gamma} = A^{1-\gamma} u(\hat{c})$.

So period t 's term in the objective is

$$\beta^t N_t u(c_t) = N_0 A_0^{1-\gamma} \underbrace{\left[\beta(1+n)(1+g)^{1-\gamma} \right]^t}_{\hat{\beta}} u(\hat{c}_t).$$

The constant $N_0 A_0^{1-\gamma} > 0$ changes no choice, so drop it: the detrended problem discounts at $\hat{\beta}$. At Step 2's numbers, $\hat{\beta} = 0.971 \times 1.0104/1.0195 = 0.963$.

At $\gamma = 1$ (Steps 1 and 3a): $\ln(A_t \hat{c}_t) = \ln A_t + \ln \hat{c}_t$, and $\ln A_t$ depends on no choice. So $\hat{\beta} = \beta(1+n)$: the same formula at $\gamma = 1$.

Why CRRA

A_t factors out of every period only because $u(A\hat{c}) = A^{1-\gamma} u(\hat{c})$. With any other utility the detrended problem still depends on t : no steady state for a grid to hold. Balanced growth needs CRRA.

The Euler equation in hats

The first-order condition of Step 3's Bellman equation is

$$G \hat{c}_t^{-\gamma} = \hat{\beta} \hat{c}_{t+1}^{-\gamma} (\alpha \hat{k}_{t+1}^{\alpha-1} + 1 - \delta),$$

and $\hat{\beta}/G = \beta(1+g)^{-\gamma}$: the Euler equation discounts by $\beta(1+g)^{-\gamma}$, not $\hat{\beta}$ (Step 4).

Step 3: The Calibrated Economy

The previous frame turned the growing economy into a stationary one, in capital per effective worker

$$\hat{k} = K/(AN):$$

$$V(\hat{k}) = \max_{\hat{k}'} \frac{\hat{c}^{1-\gamma}}{1-\gamma} + \hat{\beta} V(\hat{k}'), \quad \hat{c} = \hat{k}^\alpha + (1-\delta)\hat{k} - G\hat{k}',$$

with $G = (1+g)(1+n)$ and $\hat{\beta} = \beta(1+n)(1+g)^{1-\gamma}$.

Three changes to Step 1: G in the budget, $\hat{\beta}$ for β , and $\gamma = 2$ for $\gamma = 1$.

3a Unit-test the growth first: $\gamma = 1$, $\delta = 1$, Step 1's α, β , Step 2's g, n . The closed form survives:
 $\hat{k}' = \alpha\beta\hat{k}^\alpha/(1+g)$.

3b Solve at $\gamma = 2$ with *all* of Step 2's numbers: $\alpha, g, n, \delta, \beta$. Both: 500 points on $[0.5\hat{k}^*, 1.5\hat{k}^*]$ around *that* economy's steady state; tolerance 10^{-6} .

Checkpoint 3

3a: error below one grid step (here 0.61).

3b: the analytic steady state

$$\hat{k}^* = \left(\frac{\alpha}{(1+g)^\gamma/\beta - 1 + \delta} \right)^{\frac{1}{1-\alpha}} = 7.81$$

lies between the smallest and the largest grid point at which your policy has $\hat{k}' = \hat{k}$.

Usual pitfalls

$\hat{\beta} < 1$ or nothing converges (here 0.963). With $\gamma = 2$, $u < 0$: start from $V = 0$ anyway, and keep the penalty for infeasible $\hat{k}'$.

Step 4: Test the Model

Steps 1–3 tested your *code*. The targeted ratios come back by construction, so they cannot test the *model*. Two checks can:

- 4a A ratio you did not target.** In the model, $C/Y = 1 - (G - 1 + \delta) K/Y$: investment replaces depreciation and equips the new effective workers. Compare it with the data's consumption share (CSHCCPUSA156NRUG, 1960–2019). What is missing from the model?
- 4b The Euler residual** (3.A1, 3.A3). At each grid point, the consumption the Euler equation asks for is

$$\hat{c}^E = [\beta(1+g)^{-\gamma} \hat{c}'^{-\gamma} (\alpha \hat{k}'^{\alpha-1} + 1 - \delta)]^{-1/\gamma},$$

with $\hat{c}'$ your consumption at $\hat{k}'$. Report the largest and the mean of $|\hat{c}^E/\hat{c} - 1|$ on the $\log_{10}$ scale. How many digits does the grid buy?

Checkpoint 4

4a: C/Y is 0.757 in the model and 0.662 in the data — name the cause of the gap in one sentence.

4b: worst residual $10^{-2.00}$, mean $10^{-2.42}$.

Usual pitfalls

$\hat{k}'$ is a grid point, so read $\hat{c}'$ at its index — no interpolation. Drop ten points at each end: the policy is censored there. The Euler equation uses $\beta(1+g)^{-\gamma}$, not $\hat{\beta}$.

Technique: 3.A1 and 3.A3 (the Euler residual), 3.A4 (computing it), 3.B4 (moments the model did not target).

What Carries Over to Your Homework

Homework 1, Part A asks for the same *kinds* of step on a different economy:

- **A2** is a Version 0 test — at different parameters — with more asked of it: the stopping bound, and how the error falls as the grid is refined.
- **A3** is Step 2 at **quarterly** frequency, with the cycle as well as the long run, and with capital and depreciation data you choose and document yourself.

Part B grades every solver by Step 4's Euler residual (B1, B2) — there at capital levels *off* the grid.

What does not carry over

The numbers. HW1's model has log utility ($\gamma = 1$) and no growth, so neither the formulas for δ and β nor any value on these slides is an HW1 answer. Copying one is a way to be wrong with confidence.

What *does* carry over is the habit: a case with a known answer first, then one change at a time, each with its own check.

Takeaways

- **Start where you know the answer.** A grid VFI that matches $\alpha\beta k^\alpha$ to within a grid step is a solver you can build on.
- **Growth is a change of units.** Divide by $A_t N_t$, and the growing economy is a stationary Bellman equation with G in the budget and $\hat{\beta}$ for β . A closed form still exists to test it.
- **Calibration inverts the steady state.** Five targets give $g, n, \alpha, \delta, \beta$; the targeted ratios come back by construction, so test the model on one you did not target.
- **A grid is honest about its limits:** a band of steady states, two digits in the Euler equation.

Bridge: What You Owe, and What Comes Next

Session 3 in one sentence. A model on paper becomes a number through six decisions; we made all six badly, measured exactly how badly, named the technique that repairs each one — and you built one yourself.

- **Lab L03** repeats the Brock–Mirman part of Block A: the `GrowthModel` class, VFI, time iteration, and the closed-form check.
- **Homework 1, Part A** is out today; both parts are due October 29. It begins exactly where 1.B1 said it would — at the Brock–Mirman benchmark.

Next week (S4)

Numerical Methods I: root-finding and optimization — we open the brentq box and retire the grid search — then differentiation, approximation, quadrature, and where Π comes from. The hands-on adds a **shock** to today's economy, replaces the grid search by **optimization and interpolation**, and turns to the **business cycle** in the data.

Before then: run L03 end to end. Judd (1998), ch. 12, for the reference treatment.